

Harley Davidson Inc.



Stock Valuation and Analysis

INVESTMENTS (F21) / [FIN5401-01-F21](#)

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Introduction to the report

This report aims at collecting the financially relevant data of Harley Davidson Inc. from a variety of sources and analyse the growth, change in prices, cost of equity, average annual return, market risk, income and debt of the company and find the fair value or intrinsic value of the company in the future. We will also analyse what the future of the company looks like and if the stock is a buy or a sell for an investor.

In order to achieve the said goal, we will be preparing a spreadsheet containing historical price data of the company and S&P 500, which will help us estimate the Beta, average annual return of the market and cost of equity. It will also show us the ups and downs of the company's prices. We will also collect the earnings data, assets, debt, and shareholder's funds data to estimate the leverage ratio which in turn will show us if the company has a stable business and steady growth.

After collecting the data, we will analyse everything and calculate the cost of equity using Capital Asset Pricing Model (CAPM) and Value of the stock in 2021 to recommend a buy or sell of the stock of Harley Davidson Inc.

In addition, we also intend to provide a brief introduction to the company, its history, its market standing and competition, estimates of its earnings and dates, price and organisational information collected from various sources.

Harley Davidson Inc.



Harley-Davidson, Inc. designs, manufactures, and sells motorcycles. The Company's products include heavyweight touring, custom, and performance motorcycles, as well as a line of motorcycle parts, accessories, and general merchandise. Harley-Davidson also provides motorcycle floor planning and parts and accessories financing to its North American and European dealers.

Harley-Davidson is based in Milwaukee, WI where it houses its corporate offices. Its manufacturing facilities are in other cities in Wisconsin as well as in Pennsylvania. Outside the US, Harley's manufacturing plants are in Thailand and Brazil. In addition, its financial services corporate office is in Illinois and its wholesale and retail operations office is in Texas. The company generates about 70% of sales in the US.

Brief History

After drawing up plans for a small engine designed to fit on a regular bike frame, William S. Harley worked with Arthur and Walter Davidson to build this "motor-bicycle." They finished the first model in 1903. While the test was mostly successful, the bike was unable to climb hills without the rider providing pedalling assistance. But Harley and the Davidson brothers did not give up. They immediately began working on a new and improved machine. They created a bigger engine and loop-frame design. After completing the prototype in 1904, they entered the bike into a local motorcycle race at the State Fair Park, where it placed fourth.

By 1906, Harley and the Davidson brothers were moving out of their shed and into their first factory, which was located at the current site of Harley-Davidson headquarters. They produced 50 motorcycles during their first year in the factory – ten times more than the year before.

Within one year of building their first factory, they were already preparing to expand it with a second floor. In September 1907, Harley-Davidson officially incorporated. They produced 150 motorcycles in 1907 and began selling them to police departments.

Harley-Davidson was awarded two Army-Navy "E" Awards for Excellence in Production, in honour of their efforts during the World War II. Over 115 years, Harley-Davidson has grown from its humble beginnings to an American icon. But no matter how many things have changed over the years, one thing has always remained the same – Harley-Davidson's commitment to producing a high quality, innovative bike.

- <https://www.lasvegasharleydavidson.com/history-of-harley-davidson/>

Financial Performance

Revenue for fiscal 2020 decreased to \$4.1 billion compared from the prior year with \$5.4 billion. The decrease was mainly due to lower sales on motorcycles and related products.

The company's net income for 2020 was \$1.3 million, compared to \$423.6 million in 2019 on lower operating income both the motorcycles and Financial Services segments.

Cash held by the company at the end of fiscal year 2020 increased to \$2.5 billion. Cash provided by operations and financing activities were \$1.2 billion and \$1.4 billion, respectively. Cash used for investing activities was \$66.8 million, mainly for origination of finance receivables.

Stocks, Price and Dividend data

The Initial Public Offering of Harley Davidson was one of the America's most storied companies. This IPO occurred 1986 and was for 1.43 million shares valued between \$9 and \$11 per share. The IPO Prospectus included all the relevant information for the IPO.

The Stock Price of HOG in the New York stock exchange was rising until dec 2006 which dropped until Jan 2009. However, the stock prices presented with an upward trend from 2009. The Motorcycle company displayed a downward trend again from Jan 2014 and kept falling until April 2020, after the price of \$15.04 in April 2020, now the price is recovering and is at \$ 36.99 as on 10-01-2021.

Recent Headlines

*Premium electric-bike brand Serial 1 signals industry evolution - Business Insider
October 4, 2021 | [businessinsider.com](https://www.businessinsider.com)*

*Harley-Davidson's Stock To Gain As Company Continues Recovery?
October 1, 2021 | [forbes.com](https://www.forbes.com)*

*HARLEY-DAVIDSON, INC. TO REPORT THIRD QUARTER 2021 RESULTS - GlobeNewswire
September 30, 2021 | [globenewswire.com](https://www.globenewswire.com)*

*HARLEY-DAVIDSON, INC. TO REPORT THIRD QUARTER 2021 RESULTS ON OCTOBER 27, 2021
September 30, 2021 | finance.yahoo.com*

*Harley-Davidson Inc. stock falls Tuesday, still outperforms market - MarketWatch
September 28, 2021 | [marketwatch.com](https://www.marketwatch.com)*

Price chart, Estimates and corporate information



According to the above chart, Calculated Earnings per share for a stock of Harley Davidson Inc. is \$3.55. Price/earnings ratio is 10.08 and stock beta is 1.35. The company had a 52-week high of \$52.059 on 05/18/21 and a 52-week low of \$27.03 on 10/15/20. Also, the dividend yield of the company was 1.61%.

Earnings estimates

HOG US Equity		Alert
Harley-Davidson Inc		Periodicity Quarters Source BEst Mean Consensus (Standard) Cur USD
1) Earnings History ERN »		2) Consensus Overview EEO »
Next Announcement	(C) 10/27/21	Trailing 12M P/E 10.08
Prior Announcement	07/21/21	Forward 12M P/E 10.89
Avg/Implied 1 Day Move	10.38% / 7.9...	Est. LTG (EPS) 41.83%
		3) Analyst Coverage ANR »
		Buy/Hold/Sell 9/8/2
		Current Price 37.22
		Target Price 47.75

Here, we can see that 9/19 analysts are saying that the stock should be bought at this point, 8/19 are recommending holding the stock and 2/19 are recommending investors to sell their stock. The last dividend date was 07/21/21 and the next dividend announcement date is showing as 10/27/21.

Relative valuation

Overview	Comp Sheets	Markets	EPS Preview	Ownership	Credit	Custom	*				
Name (BICS Best Fit)	Mkt Cap (USD)	Last Px (USD)	Chg Pct 1D	Chg Pct 1M	Rev - 1 Yr Gr:Y	EPS - 1 Yr Gr:Y	P/E	ROE	Dvd 12M Yld		
Median	20.16B	30.02	1.65%	-0.45%	-11.62%	-29.02%	10.14	17.49%	1.61%		
101 HARLEY-DAVIDSON INC	5.73B	37.22	2.28%	0.32%	-24.38%	-76.98%	10.08	30.02%	1.26%		
102 HONDA MOTOR CO LTD	54.41B	30.02	1.14%	-1.09%	-11.79%	46.37%	6.06	11.17%	3.26%		
103 KAWASAKI HEAVY INDUSTRIES	3.66B	21.77	0.62%	-0.45%	-9.31%	--	176.85	3.36%	0.82%		
104 SUZUKI MOTOR CORP	21.12B	42.99	2.49%	-2.86%	-8.89%	0.54%	10.21	13.17%	1.10%		
105 PIAGGIO & C. S.P.A.	1.27B	3.56	0.92%	-2.35%	-13.65%	-30.97%	16.73	17.49%	3.61%		
106 PIERER MOBILITY AG	2.03B	89.91	1.30%	-0.13%	0.67%	-23.63%	21.81	21.77%	0.64%		
107 YAMAHA MOTOR CO LTD	9.61B	27.44	1.65%	2.33%	-11.62%	-29.02%	7.23	20.37%	3.25%		
108 GENERAL MOTORS CO	85.03B	58.57	3.77%	19.60%	-10.75%	2.61%	6.54	28.67%	--		
109 NISSAN MOTOR CO LTD	20.16B	4.77	2.13%	-6.49%	-20.41%	-66.97%	--	-2.46%	--		
110 FORD MOTOR CO	60.40B	15.12	1.54%	16.04%	-18.45%	--	7.94	23.27%	--		
111 HYUNDAI MOTOR CO	36.63B	171.35	1.99%	-1.68%	-1.65%	-53.79%	13.97	4.20%	1.95%		

Here, we can see the competition of Harley Davidson inc. in the market we have

Honda Motor co Ltd with Market Capital of \$54.41 Billion and a P/E ratio of 6.06.

Kawasaki heavy Industries with Market Capital of \$3.66 Billion and P/E ratio of 176.85.

Suzuki Motor Corp. with a Market Capital of \$21.12 Billion and a P/E ratio of 10.21.

Collection of Data

For the purpose of collection of data and estimating the fair price or stock price of Harley-Davidson, we have collected data from various sources and accumulated them in an excel workbook.

The sources used for the extraction of data are

1. Yahoo! Finance
2. Bloomberg Terminal

Types of data collected for the purpose

1. Balance sheet
2. Income statement
3. Cash flow statement
4. Monthly Stock price data from 1/1/2014 to 10/1/2021.
5. Monthly dividend data from 2/14/2014 to 6/9/2021.

Estimation of the stock price of Harley-Davidson Inc. for 2021

To calculate the cost of equity we need

Risk-free rate, Beta: (slope function) (β), Average annual return of market

$$\text{CAPM} : E(r_e) = r_f + \beta[E(r_m) - r_f]$$

From the extracted data we have found that

Risk-free rate: 0.0133

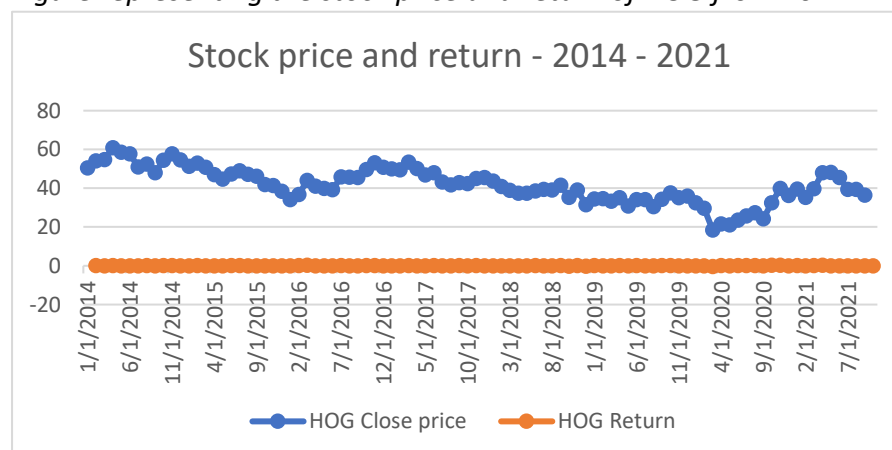
Beta: (slope function) (β): 1.432291033

Average annual return of market: 0.130750841

Therefore,

Cost of Equity (CAPM) = 0.181523786

Figure representing the stock price and return of HOG from 2014 – 2021.

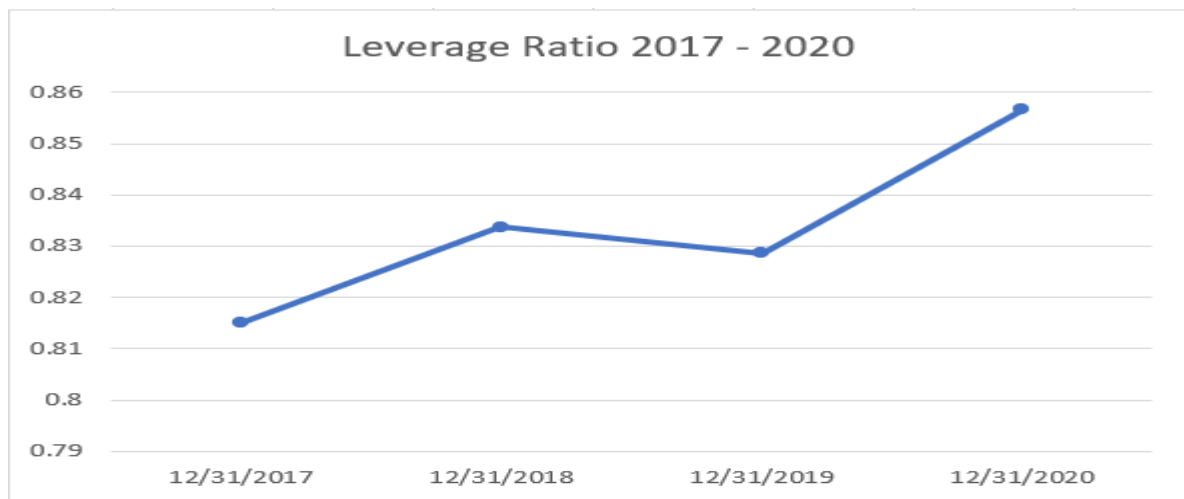


Dividend Discount Model (DDM)

We will be using Dividend Discount Model (DDM) to find the intrinsic value of the share. According to the dividend discount model, the intrinsic value of an asset equals to the present value of all future cash flows, discounted at the appropriate risk-adjusted rate of return. However, we will use Gordon Growth Model also known as One-stage DDM or Constant Growth DDM.

We are not using the two-stage model because we use Two-stage DDM to value companies with temporarily high growth, analysts use a multistage version of the dividend discount model. However, we do not have a temporary high, but we have temporary drop in dividend which is in the recovery stage now.

To apply the Gordon Growth Model to estimate the stock price, it must satisfy three assumptions. (1) stable business, (2) steady growth, and (3) stable financial leverage.



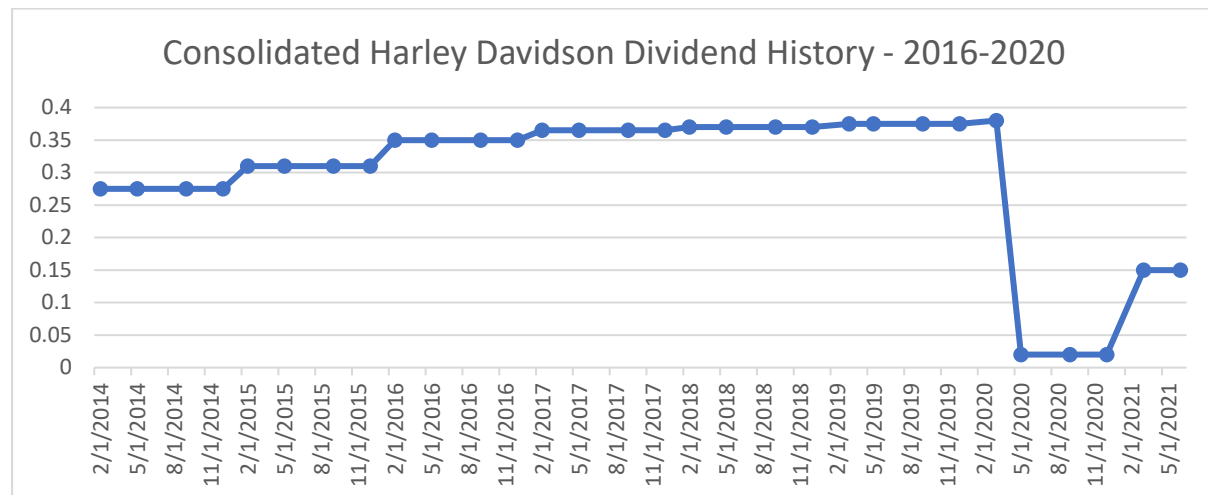
As we can see above, we have a steady growth in the leverage ratio from 2017 – 2020 which shows that the company has a steady growth and stable business. Leverage ratio is calculated by

$$\text{Leverage ratio} = \text{total debt (liabilities)} / \text{total assets}$$

The debt, assets and equity data were collected from Yahoo! finance, and we have found the following leverage ratios for the years.

12/31/2017	0.815067
12/31/2018	0.833677
12/31/2019	0.82865
12/31/2020	0.856561

We have the following dividend history for the company which shows that the company had a steady growth until 2020 and had a sharp decline in May 2020 due to covid which is an exception and not the rule. We can also see that the company is making recovery in 2021 therefore we are considering this as constant growth.



According to Bloomberg terminal, the 5 year dividend growth rate is -19.14%

Also, for the We have calculated the annual dividend growth rate by taking the formula

$(D2/D1)^{1/(D2-D1)}-1$ where

$D1$ is annual dividend for 2014 = 1.1

$D2$ is annual dividend for 2020 = 0.44

Therefore the annual dividend growth rate

$= (0.44/1.1)^{1/(2020-2014)}-1$

$= -0.14163 = -14.16\%$

We can see above that the growth rate is **-0.14163** and cost of equity according to CAPM is **0.181523786**.

Opinion on the stock

To find the Value of the stock in 2021 we need Dividend per share in 2020, Cost of equity and Annual dividend growth rate. Which are calculated and shown in the Price estimation tab of the spreadsheet.

Looking at the results of our estimation, the stock of Harley-Davidson Inc. closed at \$36.46 on Sep. 21, 2021. Compared to the intrinsic value of \$1.00 estimated from constant-growth model, the stock of Consolidated Edison Inc. is significantly undervalued. Therefore, my recommendation for the stock of Consolidated Edison Inc. is "strong sell".

Conclusion

From the above analysis, we have found that there are multiple factors affecting the fair value of a stock. The data that we collected from other sources stated that only 2/18 analysts think that the stock should be sold at this period however, we have also found that the stock should be sold because the the stock of Harley-Davidson Inc. closed at \$36.46 on Sep. 21, 2021, was higher than the intrinsic value of \$1.00 we estimated from constant-growth model, the stock of Consolidated Edison Inc. is significantly undervalued according to this model.

We also found that Harley Davidson is a constant growth company successfully growing from more than 100 years. We have learned about the history, its financial standing, potential growth, competition etc.

I have observed from the historical data and different ratios from various sources that although the company is facing harsh times at this point of time and may face even harder time in the near future, the company has the potential to recover and grow.

References

1. <https://finance.yahoo.com/>
2. Bloomberg Terminal
3. <https://www.marketbeat.com/stocks/NYSE/HOG/>
4. <https://simplywall.st/stocks/us/automobiles/nyse-hog/harley-davidson#valuation>
5. <https://en.wikipedia.org/wiki/Harley-Davidson>
6. Investments Textbook
7. Class notes and slides